

A PROJECT TITLE ON

Accident Hotspot Detection

Using DBSCAN-Based Anomaly Detection

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING • BACHELOR OF TECHNOLOGY • A.Y. 2026-27

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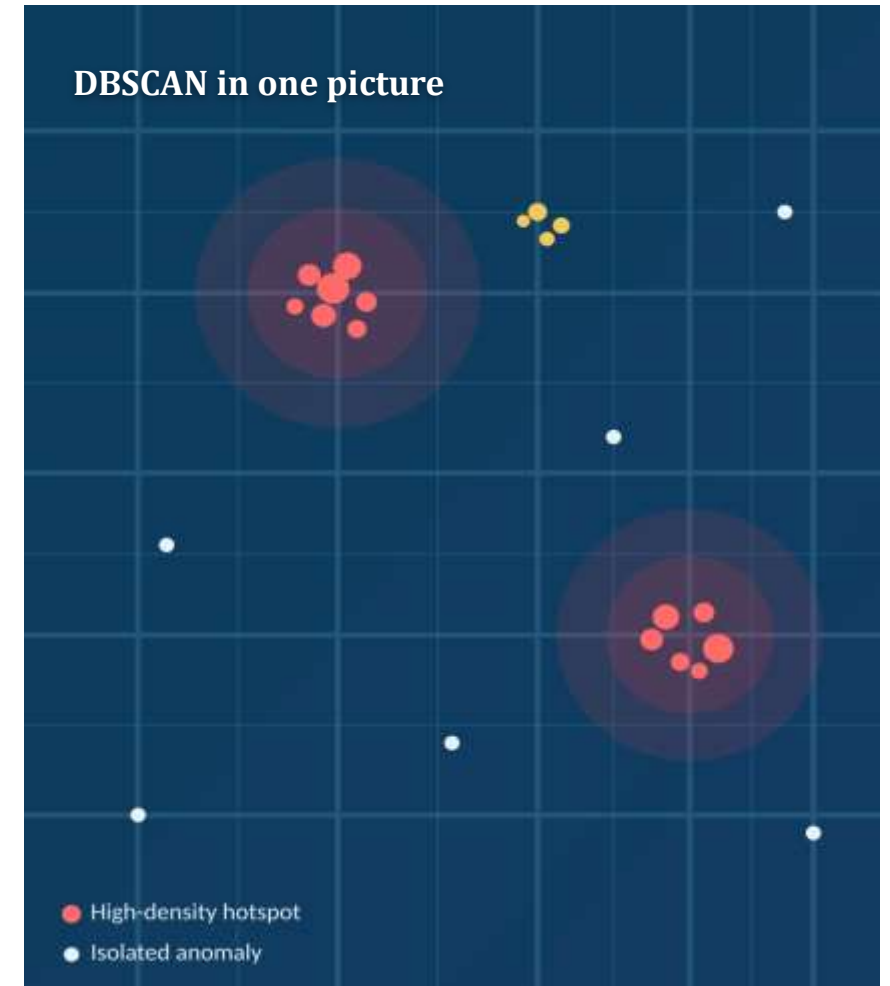
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Abstract

- Road accidents frequently recur at specific locations due to poor road conditions, congestion, sharp turns, inadequate lighting, or traffic-rule violations — but manually spotting these hotspots is slow and error-prone.
- This project builds a Machine Learning-based Accident Hotspot Detection System using DBSCAN, an unsupervised algorithm that groups accident-prone locations while flagging isolated incidents as anomalies.
- DBSCAN doesn't need the number of clusters specified in advance, handles irregularly shaped clusters, and copes well with noisy geospatial data — well suited to real-world accident records.
- Data is drawn from public sources (US Accidents, UK Road Safety, or Indian Road Accident datasets), with features like location, time, road type, speed limit, weather, and severity.
- The workflow covers data collection, preprocessing, EDA, feature selection, and DBSCAN clustering — validated using Silhouette Score, Davies-Bouldin Index, and expert review.
- Locations are classified as high, medium, low-risk, or isolated anomalies and visualized on interactive maps (Folium, Plotly) for traffic and safety planning.



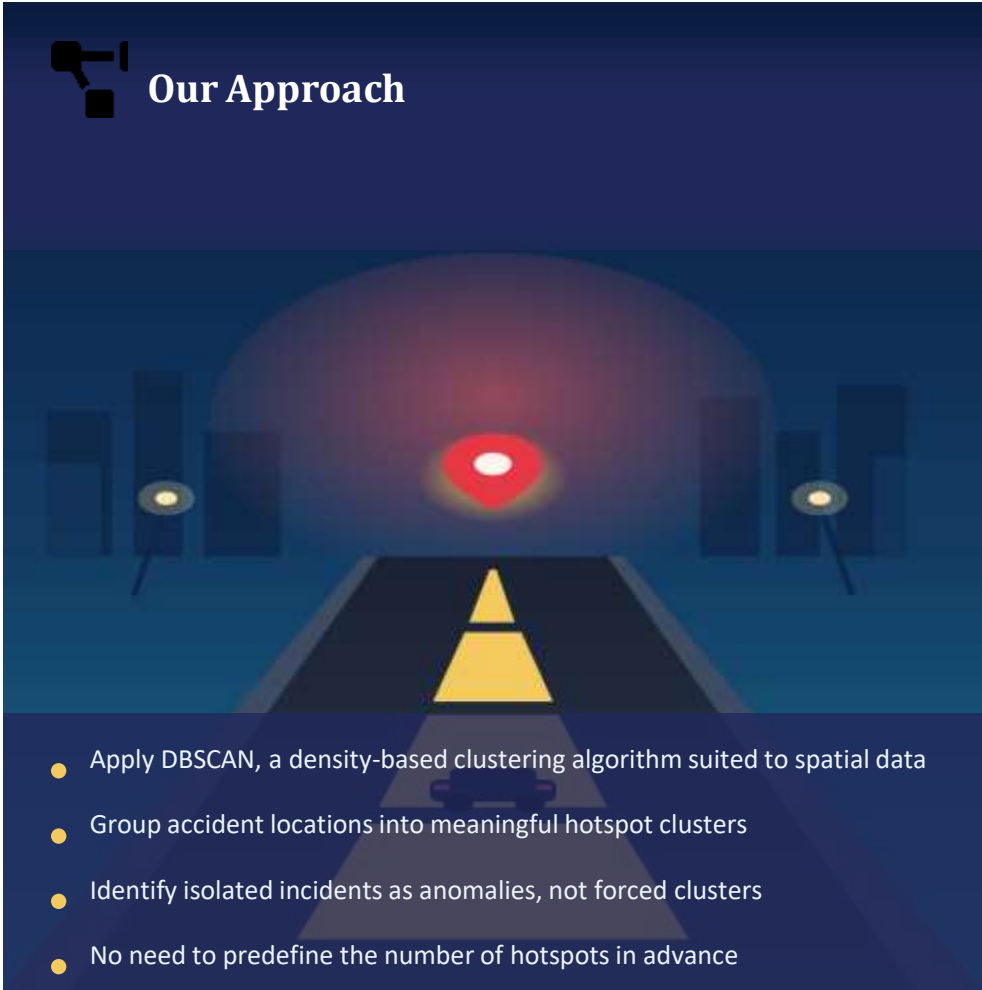
Introduction

Why this matters

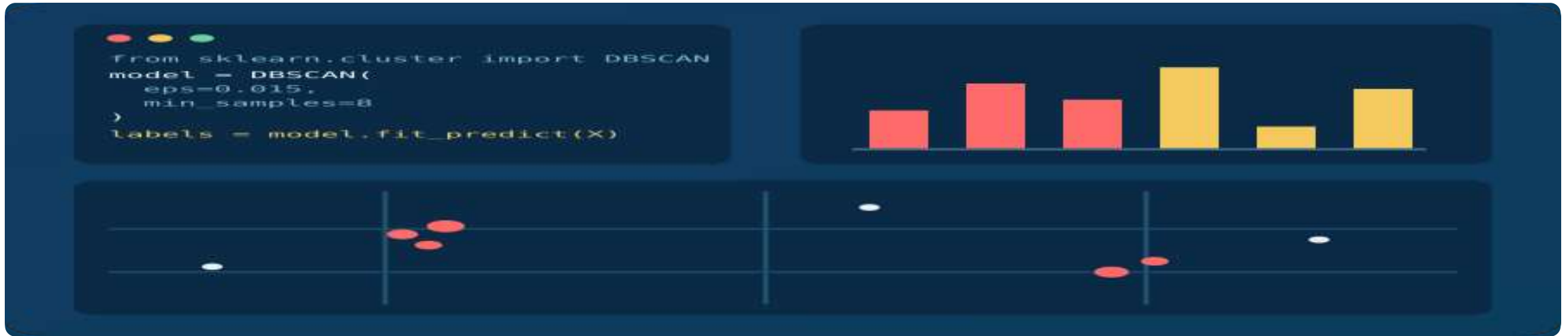
- Road traffic accidents remain a major public safety challenge, often clustering around specific locations.
- Causes include poor road design, heavy congestion, sharp turns, inadequate lighting, and frequent traffic-rule violations.
- Traditional approaches rely on manual analysis of accident reports — slow, inconsistent, and difficult to scale.
- Growing availability of geo-tagged accident records from open datasets enables automated, ML-based pattern detection.



Our Approach

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- Apply DBSCAN, a density-based clustering algorithm suited to spatial data
 - Group accident locations into meaningful hotspot clusters
 - Identify isolated incidents as anomalies, not forced clusters
 - No need to predefine the number of hotspots in advance

Software Requirements



Programming Language

Python 3.x



Libraries

Pandas, NumPy, Scikit-learn (DBSCAN), Matplotlib, Seaborn, Folium, Plotly



Development Environment

Jupyter Notebook / Google Colab / Visual Studio Code



Operating System

Windows / Linux / macOS